

Jianghui SUN

Tel: 2173903905 | E-mail: sun112@illinois.edu

EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science (MCS)

Xi'an Jiaotong-Liverpool University

Bachelor of Science in Information and Computing Science | GPA: 3.8/4.0

Urbana, IL, USA

Aug 2026 - Expected Dec 2027

Suzhou, China

Sep 2022 - Jul 2026

PUBLICATIONS

Sun, J., et al. "EmoSound: A Multimodal AI Agent Framework for Emotion-Aware Audio Accompaniment of Emoticons." *Advances in Brain-Inspired Cognitive Systems, Lecture Notes in Computer Science*, Vol. 16568, Springer, 2026. Accepted for publication; forthcoming. Presented at BICS 2025.

EXPERIENCE

Research Assistant | Xi'an Jiaotong-Liverpool University

Nov 2025 - Jan 2026

Fine-grained Emoji Understanding via DINOv2-Qwen-VL Framework

- Built a multimodal framework combining DINOv2 and Qwen-VL for fine-grained emoji understanding and description generation.
- Designed adapter-based alignment modules to map visual representations from DINOv2 into the vision feature space of Qwen-VL, enabling seamless integration with the model's multimodal architecture.
- Connected the aligned visual features to the Qwen-VL language decoder to generate detailed textual descriptions capturing subtle emoji facial expressions.

Research Assistant | Xi'an Jiaotong-Liverpool University

Sep 2025 - Nov 2025

EmoSound: An AI Agent for Emotion-Aware Audio Accompaniment of Emoticons

- Developed a multimodal AI agent for emotion-aware audio accompaniment, addressing the lack of structured emoji and emotion-aligned audio datasets for visually impaired users.
- Fine-tuned CLIP with a lightweight cross-attention soft prompt module, significantly improving emotion recognition under low-resource conditions.
- Integrated multimodal retrieval and agent-based audio generation, enabling expressive audio descriptions of visual emojis and improving emotion alignment and usability in user studies.

Machine Learning Algorithm Engineering Intern | Rockontrol Technology Group Co., Ltd.

Jun 2024 - Aug 2024

- Developed a multi-horizon air quality prediction system using ~40k+ records of air quality, carbon emission, and meteorological data.
- Built and compared RNN, LSTM, GRU, and Transformer models with sliding-window inputs (24h history -> 6h/12h/24h forecasts) for PM2.5, O3, and AQI prediction.
- Optimized models in PyTorch using RMSE/MAE, where LSTM reduced RMSE by ~12% vs baseline RNN and achieved the most stable performance across all horizons.

PROJECT EXPERIENCE

Cross-modal Audio Generation for Visually Impaired Users in Social Chat

Jun 2025 - Sep 2025

- Developed a production-oriented web application for generating emotion-aligned audio from user-uploaded images.
- Designed a training-free Multimodal RAG framework with a cross-modal knowledge graph and dual-vector databases, enabling retrieval across images, text, and audio.
- Engineered a dual-path inference pipeline (image -> tags & image -> descriptions) with Qwen-based aggregation for fine-grained emotional alignment.

Daily Reading Tracker

Mar 2025 - Jun 2025

- Developed and deployed a full-stack web application using Java for tracking and managing reading activities.
- Implemented the core reading-log module across both front-end and back-end, and contributed to the associated technical document.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL

AI/ML: PyTorch, Transformers, scikit-learn, Computer Vision, NLP, LoRA/PEFT, LLMs, RAG, LangChain

Backend & Tools: FastAPI, Spring Boot, REST APIs, Docker, Git, GitHub, Jupyter, Postman, n8n